HopStop: Stop Money-Mule Transfers at the First Hop

In-Transaction Graph-Velocity AI Scoring on IBM Z & LinuxONE with Snap ML and Telum AI Acceleration

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END-TO-END P99 LATENCY

0.202 ms

p50: 0.040 ms (batch=1) • 50x faster than <10 ms payment SLA

MINORITY-CLASS PR-AUC

0.8317

+56.7% relative lift over Velocity-Rules Baseline (0.5306)

CHAINS STOPPED ≤ HOP 2

94.0%

47/50 test laundering chains frozen before Hop 3 dispersal

SUSTAINED THROUGHPUT

109,612 TPS

Micro-batch=64 scoring (20,366 TPS single-worker stream)

EXECUTIVE SUMMARY

Mule accounts forward stolen cyber-fraud funds within minutes across instant payment rails, while bank AML checks often run hours later. **HopStop** embeds a stateful O(1) sliding-window graph feature engine and an IBM Snap ML / Tensor-Tree Gradient Boosted Decision Tree directly into the payment execution path on LinuxONE—pausing suspicious transfers at Hop 1 or Hop 2 with Top-3 analyst reason codes and tamper-evident SHA-256 audit logs.

The Golden-Minutes Gap: Stolen Money Moves in Minutes, AML Runs in Hours

WHAT BREAKS TODAY (POST-SETTLEMENT BATCH AML)

- **Rapid Multi-Hop Layering:** Organized cyber-fraud syndicates use mule accounts to split and forward stolen funds across 3–5 banks within **2 to 10 minutes** using instant rails (IMPS/UPI/ACH/Wire).
- **Hours-Late Detection:** Traditional AML transaction monitoring runs in hourly or end-of-day batch jobs—flagging the account long after it has been drained to zero.
- **High False Positives on Static Rules:** Simple velocity rules flag legitimate corporate payroll fan-out and merchant settlements, overwhelming fraud analysts.
- **1930 Helpline Recovery Bottleneck:** By the time a victim reports a scam and police issue a freeze notice, funds have scattered across dozens of downstream cash-out accounts.

THE HOPSTOP PARADIGM (IN-TRANSACTION INTERDICTION)

- **Pre-Settlement Scoring (< 1 ms):** Score every outgoing transfer *inside* the payment authorization window before money leaves the first mule account.
- **Causal Graph-Velocity Features:** Track per-account 15m/1h/24h fan-in, fan-out, pass-through ratio (money out vs. money in within minutes), and time-since-last-credit in O(1) memory lookups.
- **3-Tier Alert-Budget Policy:** Instead of blanket account freezes that hurt retail users, HopStop issues **ALLOW**, **STEP-UP** (OTP / cooling period), or **HOLD** calibrated to the bank's exact 1% analyst alert capacity.
- **Funds Preserved at Hop 1/2:** Stolen savings stay frozen in the first or second hop account so banks and cybercrime cells can recover them intact.

End-to-End Co-Located Architecture on IBM Z / LinuxONE (`s390x`)

1. Payment Stream
IBM AMLworld transactions replayed in strict timestamp order. Salted SHA-256 hashing of account IDs at ingestion so no raw PII enters logs or models.

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2. O(1) Feature Engine
In-memory sliding deques (15m, 1h, 24h) compute 12 causal features: fan-in/out, pass-through ratio, time-since-credit, rapid drain flag, Welford Z-score.

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3. Snap ML / Tensor-Tree
36-tree GBDT scorer using `tree_format=auto`: targets Telum Integrated Accelerator for AI (`libzdnn` NNPA) or branch-minimized CPU tensor-trees.

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4. Policy & Audit Layer
Calibrated to 1.0% alert budget: returns ALLOW, STEP_UP, or HOLD + Top-3 Reason Codes + SHA-256 hash-chained audit record.

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5. Carbon Analyst UI
Live queue of held transfers, Multi-Hop Chain Interdiction Graph, and real-time latency telemetry (`http://localhost:8080`).

12 ZERO-LEAKAGE CAUSAL GRAPH-VELOCITY FEATURES

Feature Group	Engineered Signals (Past Events Only)
Graph Degree Burst	`fan_in_1h`, `fan_out_1h`, `fan_in_24h`, `fan_out_24h`
Mule Pass-Through	`pass_through_ratio_15m`, `pass_through_ratio_1h`, `rapid_turnover_flag`
Temporal & Novelty	`time_since_last_credit_log`, `counterparty_novelty`, `out_tx_velocity_1h`
Amount & Rail Risk	Online Welford `amount_zscore`, `payment_format_risk` (Wire/ACH)

STRICT PREVENTION OF LABEL LEAKAGE & TRAIN/SERVE SKEW

- **Identical Online/Offline Code:** Both offline training and live REST stream scoring call `StreamingFeatureEngine.extract_and_update(tx)`.
- **Extract-Before-Update Ordering:** Features for transaction T_i are computed strictly before T_i updates sender and receiver deques.
- **Chronological Split:** Trained on Days 1–7 (35,648 tx) and evaluated on unseen future Days 8–10 (15,232 tx).

Why IBM Z / LinuxONE Is Core to the Architecture (Not Decoration)

1. CO-LOCATED IN-TRANSACTION AI

Core banking and payment switches run on IBM Z. Calling an off-platform cloud GPU/CPU cluster adds **5–30 ms of network serialization**, TLS handshakes, and tail-latency spikes—breaking payment SLAs.

Running HopStop directly on the same **IBM Z / LinuxONE (`s390x`)** platform keeps transaction state in local memory and eliminates the external network hop.

2. TELUM NNPA & IBM SNAP ML

IBM Telum (z16) and Telum II (z17) integrate an on-chip **Integrated Accelerator for AI (NNPA, Facility 165)**.

HopStop's runtime (`hopstop/ibmz_runtime.py`) probes `libzdnn.so` via `zdnn_is_nnpa_installed()` and configures **IBM Snap ML** with `tree_format="auto"`, compiling decision trees into flat tensor operations for hardware acceleration with automatic fallback to CPU tensor-trees.

3. PERVASIVE SECURITY & AUDIT

Zero Off-Platform Exposure: Sensitive payment graphs never leave the LinuxONE instance.

CPACF Hardware Crypto: Account identifiers are hashed at ingestion with salted SHA-256, and every `HOLD` / `STEP_UP` decision is linked into a tamper-evident **SHA-256 hash chain** (leveraging CPACF on `s390x`). Containerized in `Dockerfile.s390x` for **IBM Secure Execution** guests.

HARDWARE DETECTION & GRACEFUL PORTABILITY ACROSS LINUXONE VMS

Because community cloud guest VMs may or may not expose the physical NNPA facility, HopStop automatically inspects `/proc/cpuinfo` (Facilities 165 & 158) and `libzdnn.so` at startup, reporting hardware status in the live telemetry console while guaranteeing sub-millisecond inference in both accelerator and CPU tensor-tree modes.

Measured Results on IBM AMLworld Schema (Days 8–10 Unseen Test Split)

Evaluation Metric (Unseen Test Set: 15,232 tx, 50 Chains)	Velocity-Rules Baseline	HopStop (Snap ML / Tensor-Tree)	Measured Gain / SLA
PR-AUC (Minority Laundering Class)	0.5306	**0.8317**	**+56.7% relative lift**
ROC-AUC	0.9035	**0.9854**	**+9.1% lift**
Laundering Recall @ Fixed 1.0% Alert Rate	44.5%	**53.3%**	**+19.8% relative lift**
Laundering Recall @ Fixed 2.0% Alert Rate	62.9%	**72.8%**	**+15.7% relative lift**
Multi-Hop Chains Stopped by Hop 2 (≤2 Hops)	86.0% (43/50)	**94.0% (47/50)**	**+8.0 pts more chains frozen**
End-to-End Decision Latency (batch=1)	—	**p50: 0.040 ms / p99: 0.202 ms**	**50x below 10 ms target**
Throughput (Single Stream / Micro-Batch=64)	—	**20,366 TPS / 109,612 TPS**	**Core payment scale**

WHY HOPSTOP BEATS STATIC VELOCITY RULES

- **Disentangles Benign Bursts from Mule Layering:** Corporate payroll fan-outs have high `fan_out_1h`, which triggers false positives in static velocity rules. HopStop combines `pass_through_ratio_15m` (22.9% importance weight), `time_since_last_credit_log` (17.4%), and `counterparty_novelty` (13.9%) to isolate true rapid pass-through mules.
- **Early Chain Interdiction Across All 8 Typologies:** Interrupts **94.0%** of multi-hop laundering chains (Fan-Out, Gather-Scatter, Scatter-Gather, Simple-Cycle, Fan-In, Bipartite, Stack, Random-Walk) at Hop 1 or Hop 2—preventing downstream dispersal to cash-out accounts.

Dataset: IBM Transactions for Anti-Money Laundering (AMLworld schema, Altman et al., NeurIPS 2023, CDLA-Sharing-1.0)

Reproducible via `python -m hopstop.train_and_benchmark`

Who Benefits & What Value HopStop Delivers

1. FRAUD OPS & RISK TEAMS

Fraud Operations Analysts: Receive a live queue of held/step-up payments with Top-3 human-readable reason codes instead of next-day batch alerts.

Bank Risk Heads: Dial the alert budget (e.g., 1.0%) to match analyst headcount while capturing +19.8% more laundering transfers than static rules.

2. LAW ENFORCEMENT & REGULATORS

1930 Cybercrime Helpline & Police Cyber Cells: Because 94% of mule chains are paused by Hop 2, stolen victim funds are still sitting in the account when the freeze notice arrives.

AML Compliance Officers (FIU-IND): Get time-stamped, SHA-256 hash-chained audit logs with reason codes for Suspicious Transaction Reports (STRs).

3. CORE BANKING & RETAIL USERS

IBM Z / LinuxONE Platform Teams: Deploy a containerized (`s390x`) scoring engine achieving 0.202 ms p99 latency with zero external data egress.

Retail Banking Customers: Protected from losing life savings to cyber scams, while middle-risk transfers trigger lightweight `STEP_UP` verification instead of wrongful account freezes.

REPRODUCIBLE OPEN-SOURCE REPOSITORY & LIVE DEMO VERIFICATION

1-Command Benchmark: `python -m hopstop.train_and_benchmark` | **1-Command Live Console:** `python -m hopstop.server` (opens at `http://127.0.0.1:8080`) | **LinuxONE Container:** `docker build -f Dockerfile.s390x -t hopstop-ibmz .`

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